

DATA266 Lab 1 — Team 5

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Course: DATA266

Team: 5

Members: Sneha Singh · Ritika Mukesh Neema

Repo: https://github.com/snehas-SJSU/Data_266-Lab1

This file is not the finished team report yet. Only Sneha Singh's results are filled in. Ritika Mukesh Neema's sections are blank. Once she adds her models, metrics, plots, and failure notes, we will merge both into one combined report and export the PDF again. Until then, every number and every image below is from one person.

Who built what: I trained my own model on every part and wrote only under `sneha_singh/`. Ritika trains her own models under `ritika_mukesh_neema/`. We do not share architecture or hyperparameters. The repo is https://github.com/snehas-SJSU/Data_266-Lab1.

How we split the work

Part	Sneha Singh	Ritika Mukesh Neema
1 — GPT (TinyStories)	My model, full GPU run	Her model
2 — Yelp polarity	My three models, 100k run	Her three models
3 — CycleGAN	My train, local FID, my audit column	Her train and her audit column
Report / Drive / Kaggle	My half	Her half

We share the raw data. I only write under `sneha_singh/`. She only writes under `ritika_mukesh_neema/`.

Part 1 — GPT from scratch

1.A Sneha Singh

I trained a small character-level GPT on TinyStories. The attention code is mine. I did not use `nn.Transformer` or `nn.MultiheadAttention`.

The tokenizer is a simple `char_to_idx / idx_to_char` map, not BPE. The model has 4 layers, 4 heads, embedding size 256, and a context of 128 characters. A causal mask stops each position from seeing the future. I tied the token weights the usual way and trained with AdamW. Bias and LayerNorm do not get weight decay.

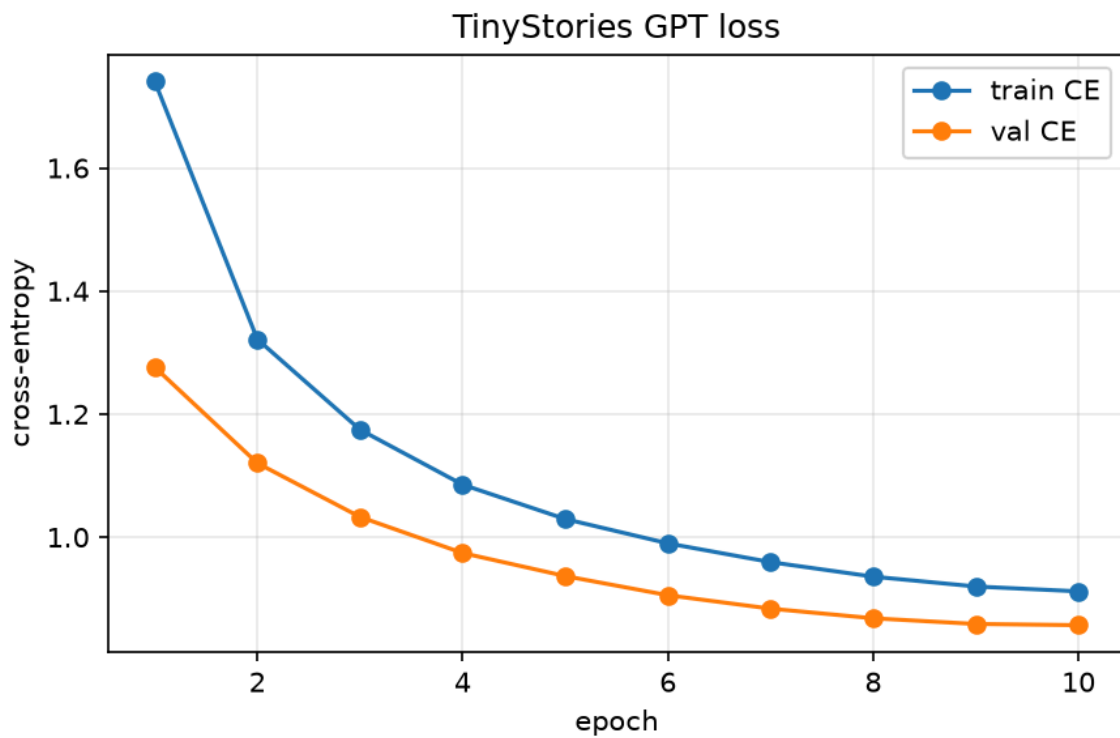
The reported run is the full one, not the old 2-epoch smoke test. I took my own split from `TinyStories-train.txt`: 100,000 train windows and 10,000 validation windows, seed 670. That text file is about 1.92 GB, so it is on Drive and not in git: https://drive.google.com/file/d/1K8wGXMKDaLaVdv7ISMNihMH33pGig-CN/view?usp=share_link

I trained for 10 epochs on an NVIDIA GeForce RTX 5090 with mixed precision. Learning rate was 0.001, with 100 warmup steps and then cosine decay. Batch size was 32. The run took 457 seconds and peaked at about 349 MB. There were no NaN losses. The checkpoint `best.pt` is 12 MB, so it is on git.

Checkpoint id: `task1_llm/sneha_singh/checkpoints/best.pt`. Log: `reproducibility/raw_logs/sneha_singh/task1_llm/train_full.log`.

Metric	Value
Train cross-entropy	0.912
Val cross-entropy	0.857
Perplexity	2.36
Bits per character	1.236
Generalization gap (val – train)	−0.055
Top-1 next-character accuracy	0.729
Distinct-1 / 2 / 3	0.225 / 0.655 / 0.856
Repeated 4-gram rate	0.120
Max gradient norm / NaN count	5.42 / 0
Parameters	3.25M
Train tokens/sec	280,325
Generation tokens/sec	409
Peak memory / train time	349 MB / 457 s

The loss falls smoothly and the validation curve stays close to the training curve. That is what I wanted from 10 epochs on this size of model.



Part 1 training and validation loss

The prompt I used for samples was “some changes. She added some nice colors”.

Greedy decoding writes a readable start, then gets stuck: “You are very happy. You are very happy.” It also doubles a word, “them them”. Temperature 0.8 does not loop as hard, but the story wanders from colors to a box, a door, and then a tree. I expected some of this. The model only sees 128 characters at a time, and it has no idea where a word ends. The three cases are written up in [task1_llm/sneha_singh/failure_analysis.md](#).

1.B Ritika Mukesh Neema

Ritika: architecture, hardware, metrics, a loss plot, a sample, and three failure cases from `task1_llm/ritika_mukesh_neema/`.

1.C Comparison

Fill this after Ritika's Part 1 is in. Do not compare until both numbers are real.

	Sneha	Ritika
Architecture	4 layers, 4 heads, emb 256, context 128, char tokenizer, causal mask	
Hyperparameters	10 epochs, batch 32, lr 0.001, warmup 100, seed 670, AdamW	
Train / val CE	0.912 / 0.857	
Perplexity / bits per char	2.36 / 1.236	
Gen. gap / top-1 acc	-0.055 / 0.729	
Distinct-1/2/3	0.225 / 0.655 / 0.856	
Repeated 4-gram / NaNs	0.120 / 0	
Params / time / peak MB	3.25M / 457 s / 349	
Joint note	Write this with Ritika: strengths, weaknesses, and what we would try next.	

Part 2 — Yelp polarity

2.A Sneha Singh

This is Yelp polarity, not IMDB. I learned the embeddings from scratch. I did not use a pretrained language model.

The Yelp files are not on Drive and not in git. The notebook downloads fancyzhx/yelp_polarity from Hugging Face when the run starts, so there is nothing for me to upload.

I wanted one simple model and two that can use word order, so a strong baseline would be obvious if the fancier models did not beat it.

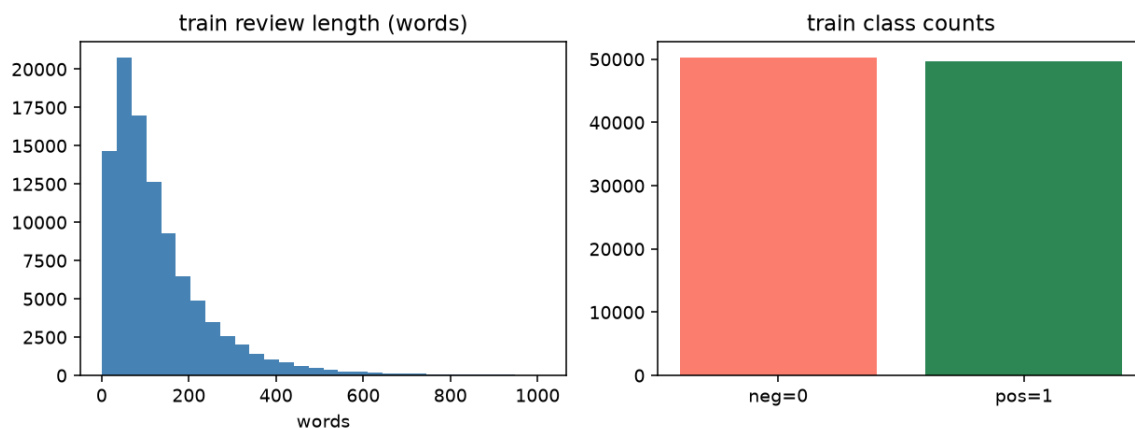
1. **Baseline.** Mean pool over the embeddings, then a linear layer.
2. **BiLSTM.** Reads the review in both directions, so negation and longer sentences have a chance.
3. **TextCNN.** Filters of width 3, 4, and 5, meant to catch short phrases like “not good”.

I lowercased, stripped punctuation, removed stopwords, and lemmatized. The vocabulary comes from the training text only. I sampled 100,000 training reviews. Eleven were empty after cleaning, so the split is 99,989 / 10,000 / 10,000. Five epochs, batch 64, embedding size 100, maximum length 128. All three models ran on my Mac: Apple M4, 16 GB unified memory, MPS. The CUDA peak-memory counter stays 0 on MPS, so I left that field at 0.0 instead of inventing a number.

Model	Acc	Macro-F1	ROC-AUC	MCC	Brier	ECE	Time
Baseline	0.923	0.923	0.973	0.845	0.059	0.011	114 s
BiLSTM	0.918	0.918	0.975	0.837	0.060	0.017	1706 s
TextCNN	0.921	0.921	0.975	0.841	0.061	0.024	108 s

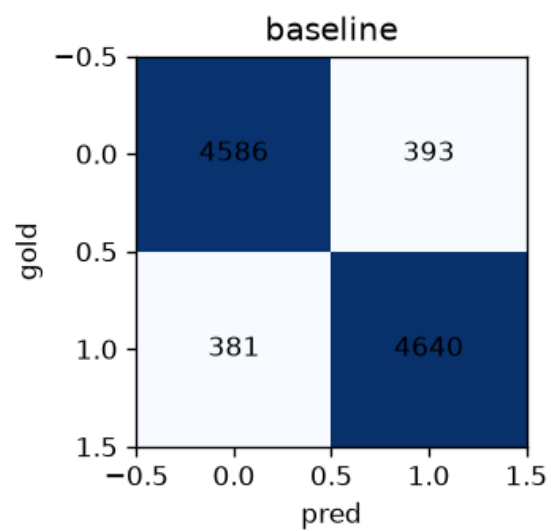
The baseline still has the best macro-F1. McNemar against the baseline is $p = 0.091$ for the BiLSTM and $p = 0.424$ for the TextCNN. On this 10k test set I cannot say the other two are clearly different. Short reviews are a bit easier than long ones for every model. Long-review macro-F1 is 0.917, 0.912, and 0.915 for baseline, BiLSTM, and TextCNN.

The length and class balance of the sample:

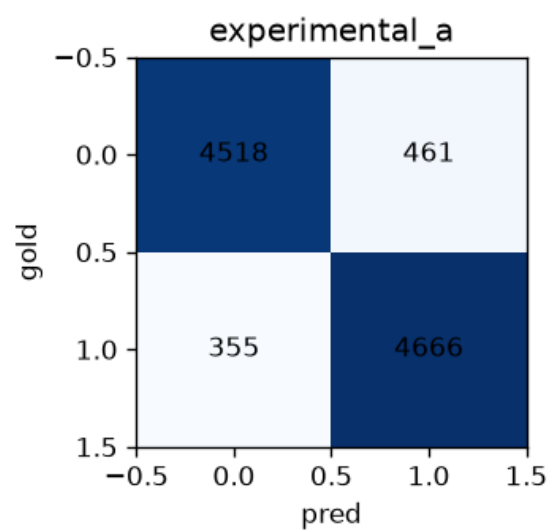


Yelp length and class balance

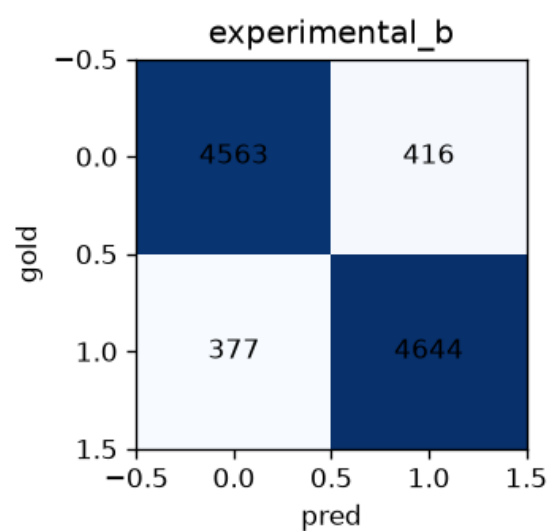
Confusion matrices on the test set:



Baseline confusion matrix



BiLSTM confusion matrix



TextCNN confusion matrix

I read 20 baseline mistakes. Five confident false positives, five confident false negatives, five near the decision threshold, and five long-review misses. The full list, with an error type and a testable fix on each row, is in `task2_sentiment/sneha_singh/failure_analysis.md`. Four of them:

- Confident false positive, gold negative: “food always good”.
- Confident false negative, gold positive: “not restaurant closed”.
- Near the threshold: “disappointed went early morning ... coffee warm not good”.
- Long review, gold positive, predicted negative: a long airport note that mixes praise with complaints.

The ones that show up most are tiny blurbs the model treats as positive, negation or mixed wording, and long reviews where the stars and the sentences pull apart.

This is 100k reviews out of about 560k, not the whole Yelp train file. Cutting reviews at 128 tokens hurts the long ones. I kept negation words, and mixed reviews are still hard.

2.B Ritika Mukesh Neema

Ritika: her three models, metrics, plots, and error notes from task2_sentiment/ritika_mukesh_neema/.

2.C Comparison

Fill this after Ritika's Part 2 is in.

	Sneha baseline	Sneha BiLSTM	Sneha TextCNN	Ritika
Architecture	Mean-pool + linear	Bidirectional LSTM	CNN widths 3, 4, 5	
Hyperparameters	emb 100, max len 128, batch 64, 5 epochs, lr from config, Apple M4 MPS	same training setup	same training setup	
Accuracy	0.923	0.918	0.921	
F1 macro / micro / weighted	0.923 / 0.923 / 0.923	0.918 / 0.918 / 0.918	0.921 / 0.921 / 0.921	
ROC-AUC / PR-AUC	0.973 / 0.973	0.975 / 0.975	0.975 / 0.975	
MCC / Brier / ECE	0.845 / 0.059 / 0.011	0.837 / 0.060 / 0.017	0.841 / 0.061 / 0.024	
Acc 95% CI	0.917–0.928	0.912–0.923	0.915–0.926	
McNemar p vs baseline	—	0.091	0.424	
Macro-F1 short / long	0.925 / 0.917	0.922 / 0.912	0.924 / 0.915	
Params / time / peak MB	4.76M / 114 s / 0.0 (MPS)	4.85M / 1706 s / 0.0	4.84M / 108 s / 0.0	
Checkpoint	checkpoints/baseline_full.pt	checkpoints/experimental_a_full.pt	checkpoints/experimental_b_full.pt	
Joint note	Write this with Ritika. My mean-pool baseline still leads. McNemar does not separate the other two on this test set.			

Part 3 — CycleGAN, photo to Monet

3.A Sneha Singh

I trained an unpaired CycleGAN. The Kaggle direction is photo to Monet. The generators are mine. I did not use a pretrained model to make the images. Inception and the other nets in the eval script are only for measuring.

Each generator is a ResNet with 9 blocks at 256×256 , reflection padding, instance norm, and a tanh output. Upsampling is nearest-neighbor times two, then a stride-1 convolution, not a transposed convolution. Each discriminator is a PatchGAN trained with least-squares loss. The losses are adversarial, cycle L1 with $\lambda = 10$, and identity at half of that. Real labels are smoothed to 0.9. An image pool of 50 feeds the discriminator. Batch size is 4. I trained 40 epochs at a constant learning rate and 40 more with decay, and each epoch resamples about 800 photos. The shorter Monet loader is cycled so those photos actually get used. Scoring uses all 7,038 photos in the photo-to-Monet direction and all 300 Monet paintings in the other direction.

Training was on a CUDA GPU with mixed precision. It took about 4.3 hours and peaked near 6568 MB. The log records device=cuda and does not print the GPU name. best.pt is 113 MB, over GitHub's limit, so it is on Drive: https://drive.google.com/file/d/1TVDCjIGkr5Xt5nrrOmkjhKA3G9ppj-2T/view?usp=share_link

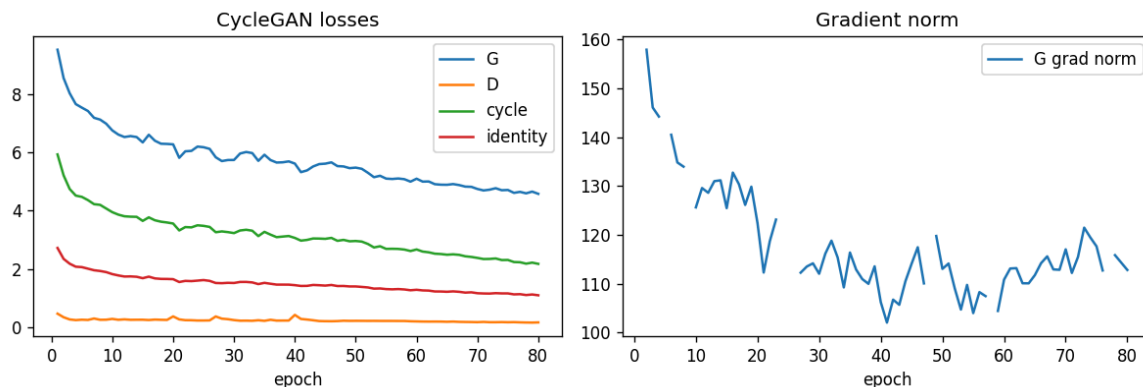
The Monet and photo folders are in one Drive folder:

https://drive.google.com/drive/folders/1BXYfhW8uZ6umK1TZFW8Un62mZVK72L7Y?usp=share_link

Direction	FID	KID	MiFID	LPIPS	Content cosine	Cycle L1
Photo → Monet	89.99	0.018	0.403	0.452	0.532	0.112
Monet → photo	92.23	0.030	0.423	0.365	0.586	0.103

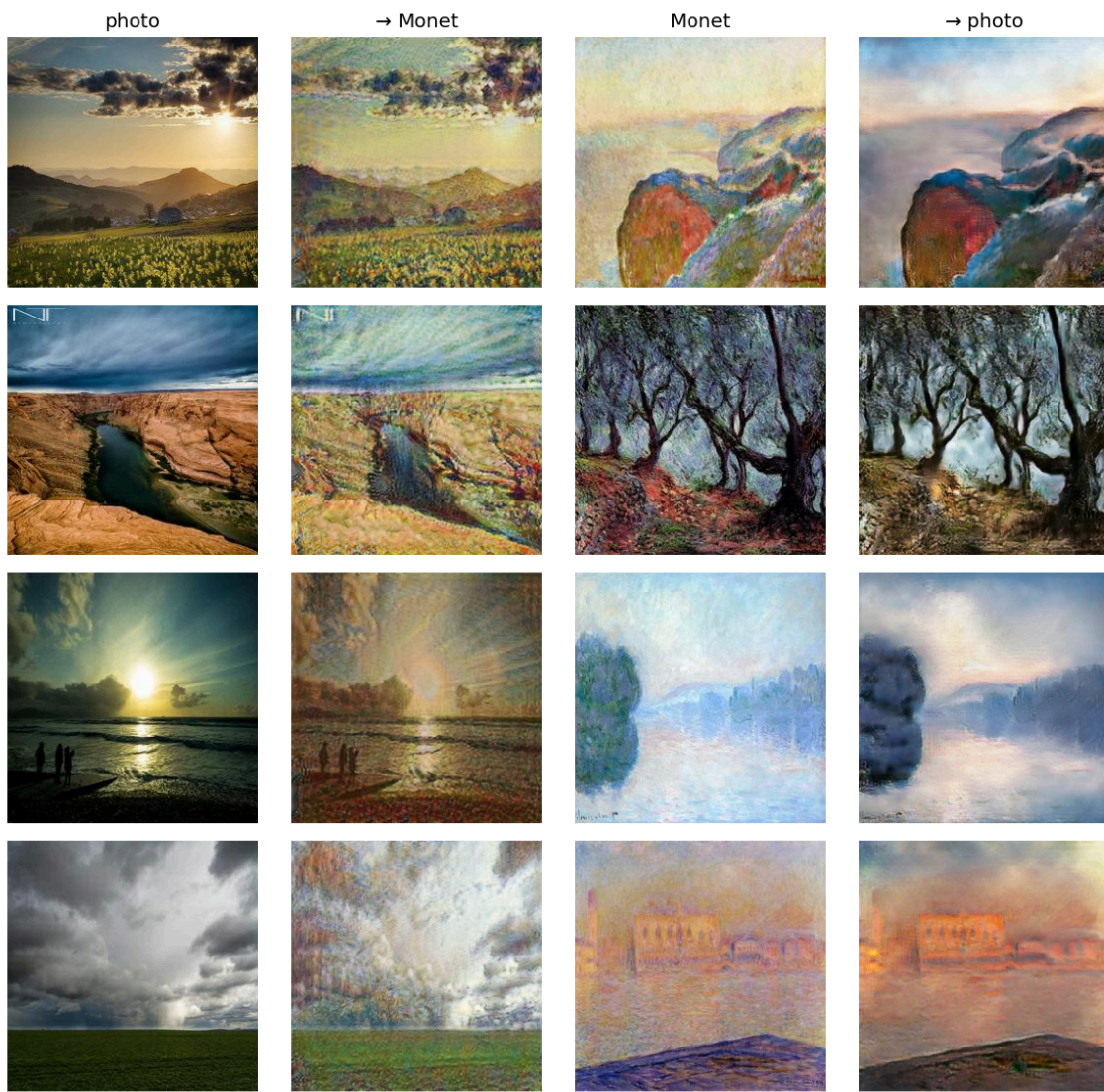
FID is the Fréchet distance between Inception features of the generated images and the real target images. MiFID here is the average cosine distance of those features after an equal subsample. Lower is better on both. My local combined score for the Kaggle direction is $(89.99 + 0.403) / 2 = 45.20$. The same two numbers are in task3_gan/sneha_singh/submission.csv. I have not uploaded that file yet, so public score, private score, and rank are still empty.

Losses fell across the 80 epochs. The discriminator loss ended near 0.17, which is low. The generator is not winning that fight, and I think that is why the brush texture gets noisy.



CycleGAN losses and gradient norm

The grid is four photos, the Monet version of each, four real Monets, and the photo version of those.



Photo, generated Monet, real Monet, generated photo

What I see in that grid, and in the 30 images I audited: the layout of the photo usually survives. Skies and water pick up a grainy, repeated dab texture. Some colors wash out. A few Monet-to-photo frames go soft or pick up a dark blob. I did not mark any of my 30 as a copied training Monet. My rough averages on the sheet were style 1.4, content 1.7, artifacts 1.4, on a 0–2 scale where higher means worse. The sheet is `task3_gan/sneha_singh/outputs/human_audit/audit_30.csv`. Agreement with Ritika is not computed yet, because she has not scored the same 30.

3.B Ritika Mukesh Neema

Ritika: her CycleGAN, hardware, FID and MiFID, a grid, and her column on the same 30 images. Folder: task3_gan/ritika_mukesh_neema/.

3.C Comparison

Fill this after Ritika's Part 3 is in. Kappa needs both of us on the same 30 files.

	Sneha A2B (photo → Monet)	Sneha B2A	Ritika
Architecture	ResNet-9, nearest upsample, PatchGAN, LSGAN, λ_{cycle} 10, identity 5, label smooth 0.9	same model	
Hyperparameters	40 + 40 epochs, batch 4, ~800 photos/epoch, pool 50, CUDA AMP	same run	
FID / KID / MiFID	89.99 / 0.018 / 0.403	92.23 / 0.030 / 0.423	
Precision / recall	0.540 / 0.293	0.660 / 0.089	
Cycle L1 / LPIPS / content cos	0.112 / 0.452 / 0.532	0.103 / 0.365 / 0.586	
G / D / cycle / identity loss	4.574 / 0.168 / 2.174 / 1.097	same training losses	
Grad norm / NaNs	112.8 / 0	same	
Params / time / images/sec / peak MB	28.3M / 15342 s / 8.34 / 6568	same	
Local (FID + MiFID) / 2	45.20	46.33	
Kaggle public / private / rank	not uploaded yet		
Human audit / κ	style 1.4, content 1.7, artifacts 1.4 (0–2, higher worse). κ pending		
Checkpoint	Drive best.pt (113 MB), not on git		
Joint note	Write this with Ritika after she scores the same 30 images.		

What is still open

1. Upload my submission.csv to Kaggle for PairProgramming_Team_5, then paste the public score, private score, and rank into this file.
2. Ritika's three sections, and the three comparison tables above.
3. This PDF is the current export. Regenerate it after Ritika's sections and the Kaggle score are filled in.

The plots in this file are the png files already stored under each task folder. They are not copied again into report/.

Papers

Vaswani et al., Attention Is All You Need (2017). The causal multi-head attention in Part 1 follows that design, written by hand.

Eldan and Li, TinyStories (2023). Part 1 is trained on that dataset.

Zhu et al., Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks (2017). Part 3 is that CycleGAN setup: two generators, two discriminators, cycle loss, and identity loss.